

Micro IV 2026 Final Exam

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July 20, 2026

- Time: 90 min (9:00-10:30).

90 minutes

- There are 3 big problems.
- You should write down the details for each question.
- Good Luck!

Problem 1 (30 points)

Let $X = \{a, b, c, d\}$. Suppose that four voters have the following preference orderings:

$$\succ_1 : a \succ b \succ c \succ d,$$

$$\succ_2 : b \succ c \succ d \succ a,$$

$$\succ_3 : c \succ d \succ a \succ b,$$

$$\succ_4 : a \succ c \succ b \succ d.$$

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For every nonempty subset $B \subseteq X$, restrict each voter's preference ordering to B . Under the Borda rule on B , an alternative ranked k -th from the bottom receives $k - 1$ points. Define the *Borda choice correspondence* C by

$$C(B) = \{x \in B \mid x \text{ receives the highest total Borda score in } B\}.$$

A choice correspondence C is said to be *rationalizable* if there exists a complete and transitive preference relation R on X such that

$$C(B) = \max(B, R)$$

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for every nonempty subset $B \subseteq X$.

1. Compute the total Borda score of each alternative when the feasible set is X , and determine $C(X)$.
2. Compute the following choice sets:

$$C(\{a, b, c\}), \quad C(\{a, c, d\}), \quad C(\{a, c\}).$$

3. A choice correspondence satisfies *Sen's property α* if, for every $A, B \subseteq X$,

$$x \in C(A), \quad x \in B \subseteq A \implies x \in C(B).$$

Determine whether the Borda choice correspondence C satisfies Sen's property α .

4. Show that every rationalizable choice correspondence satisfies Sen's property α . Hence, determine whether the Borda choice correspondence C is rationalizable.

Problem 2 (30 points)

Let $\mathcal{U}$ be the set of all finite utility vectors. Consider the generalized critical-level utilitarian social welfare ordering R defined as follows: there exists a strictly increasing function $g : \mathbb{R} \rightarrow \mathbb{R}$ with $g(0) = 0$ and a critical utility level $\gamma \in \mathbb{R}$ such that, for any $u_N = (u_i)_{i \in N}$ and $v_M = (v_i)_{i \in M}$,

$$u_N R v_M \iff \sum_{i \in N} (g(u_i) - g(\gamma)) \geq \sum_{i \in M} (g(v_i) - g(\gamma)).$$

For $a > 0$, let $A_n(a) = (\underbrace{a, \dots, a}_{n \text{ individuals}})$ denote a population of n individuals, each of whom has utility a .

1. For $a, b > 0$ and positive integers n, m , provide a necessary and sufficient condition for $A_n(a) R A_m(b)$.
2. Fix $a > b > 0$ and a positive integer n . Provide a necessary and sufficient condition under which

$$A_m(b) P A_n(a)$$

holds for all sufficiently large positive integers m . Express your condition in terms of b and γ .

3. We say that R satisfies the *repugnant conclusion* if, for every $a > 0$, every positive integer n , and every $b \in (0, a)$, there exists a positive integer m such that

$$A_m(b) P A_n(a).$$

Provide a necessary and sufficient condition on the critical utility level γ under which R avoids the repugnant conclusion.

4. We say that R satisfies the *sadistic conclusion* if there exist utility vectors $u_N \in \mathbb{R}_{++}^N$, $v_M \in \mathbb{R}_{--}^M$, and a background population w_K such that

$$(v_M, w_K) P (u_N, w_K).$$

Provide a necessary and sufficient condition on the critical utility level γ for avoiding the sadistic conclusion.

Problem 3 (40 points)

Let $N = \{1, 2, 3\}$ be the set of players and $A = \{0, a, b\}$ be the set of alternatives, where 0 denotes the status quo. The implementation costs are given by $c(0) = 0, c(a) = 9, c(b) = 6$. The players' valuation functions are given by

	0	a	b
1	0	7	3
2	0	5	4
3	0	1	6

The Clarke mechanism chooses an alternative

$$x(v) \in \arg \max_{x \in A} \left\{ \sum_{i \in N} v_i(x) - c(x) \right\},$$

and each player's payment is given by

$$t_i(v) = \frac{c(x(v))}{3} - \left(\sum_{j \neq i} v_j(x(v)) - \frac{2}{3}c(x(v)) \right) + \max_{x \in A} \left(\sum_{j \neq i} v_j(x) - \frac{2}{3}c(x) \right).$$

1. Compute the social surplus

$$\sum_{i \in N} v_i(x) - c(x)$$

for each alternative $x \in A$, and determine the alternative selected by the Clarke mechanism.

2. Compute the Clarke payment $t_i(v)$ for each player $i \in N$.

3. Verify that

$$\sum_{i \in N} t_i(v) \geq c(x(v)).$$

Determine whether equality holds in this example.

4. Suppose that player 1 reports the valuation

$$v'_1(0) = 0, \quad v'_1(a) = 8, \quad v'_1(b) = 0,$$

while the other players report truthfully.

Determine the alternative selected by the Clarke mechanism and player 1's payment. Then compare player 1's utility

$$u_1 = v_1(x) - t_1$$

under truthful reporting and under this misreport.